Examiner only

7. A student investigated the reaction between hydrochloric acid (HCl) and calcium carbonate ($CaCO_3$). She measured the volume of gas produced for different concentrations of acid after 60 seconds.

She started each reaction at the same temperature, with the same mass of calcium carbonate and the same volume of acid. The acid was in excess each time.

The table below shows the time taken to produce 60 cm^3 of CO_2 for each concentration of HCl.

Concentration of HCl (mol/dm^3)	Time to produce 60 cm^3 of CO_2 (s)	Mean rate of reaction (cm^3/s)
0.5	234	0.26
1.0	118	0.51
2.0	58	....

(a) (i) **Complete** the table to two significant figures. [2]

(ii) Estimate the time taken to produce 60 cm^3 of carbon dioxide if a concentration of 4.0 mol/dm^3 HCl is used. [1]

time = ... s

Examiner only

(b) (i) State and explain, in terms of particles, the effect of increasing acid concentration on the rate of reaction. [3]

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(ii) The student noticed that when the experiments ran for 10 minutes, she obtained the same volume of carbon dioxide in each case.

Explain her observation. [2]

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END OF PAPER

THE PERIODIC TABLE

1	2					Group						3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

Key	
A_r	relative atomic mass
Symbol	
Name	
Z	atomic number